

FOURTH SEMESTER DIPLOMA EXAMINATION IN INSTRUMENTATION ENGINEERING

INDUSTRIAL INSTRUMENTS I

MODEL QUESTION PAPER

(Maximum marks: 100)

(Time: 3hrs)

PART-A

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. List four units of pressure.
2. List the materials used for the construction of bellows.
3. Mention the advantages of air purge method of level measurement.
4. List the merits of pyrometers.
5. State Seebeck effect.

(5x2=10)

PART- B

II Answer any *five* of the following. Each question carries 6 marks.

1. Compare different types of pressure.
2. Explain the working of ultrasonic level gauge.
3. Describe level switch.
4. Describe different temperature scales.
5. Illustrate the working of vapour pressure thermometer.
6. Explain any two thermistor characteristics.
7. State the laws of thermocouple

(5x6=30)

PART- C

(Answer one full question from each module. Each question carries 15 marks)

MODULE-I

III (a) Explain the working of bell type manometer. (7)

(b) Describe the working of Mcleod gauge. (8)

OR

IV (a) Explain the method of pressure measurement using diaphragms. (8)

(b) Illustrate the working of fiber optic pressure sensor. (7)

MODULE-II

- V (a) Illustrate the operation of torque tube level indicator. (8)
(b) Describe radiation absorption method of level measurement. (7)

OR

- VI (a) Describe the construction and working of capacitive type level indicator. (7)
(b) Explain the working of level transmitter . (8)

MODULE -III

- VII (a) Convert to Kelvin scale. (1) 20°C (ii) 130°C (iii) 220°F (iv) 125°F (8)
(b) Explain the construction and working of radiation pyrometer. (7)

OR

- VIII (a) Explain the working of bimetallic thermometer. (7)
(b) Describe the construction and working of optical pyrometer. (8)

MODULE-IV

- IX (a) Explain the construction and working of RTD (8)
(b) State and explain the laws of thermocouple. (7)

OR

- X (a) Describe thermistor and its types. (8)
(b) Explain the working of thermocouple and its types. (7)